

ISLAMIC UNIVERSITY OF TECHNOLOGY



Course No : **EEE 4308**
Course Title : **Electrical Circuit II Lab**
Name of the Project : **SAP-1 (ARCHITECTURE-BASED 6-BIT COMPUTER)**

DESIGN AND IMPLEMENTATION (PHASE C, D,E)

Date of Submission : **.11.2023** Group Name : **N.H.P.**
Department : **EEE**

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Objective: The objective of this project is to build a 6-bit computer based on the SAP-1 (Simple-As-Possible) architecture using combinational and sequential circuits.

Block Diagram:

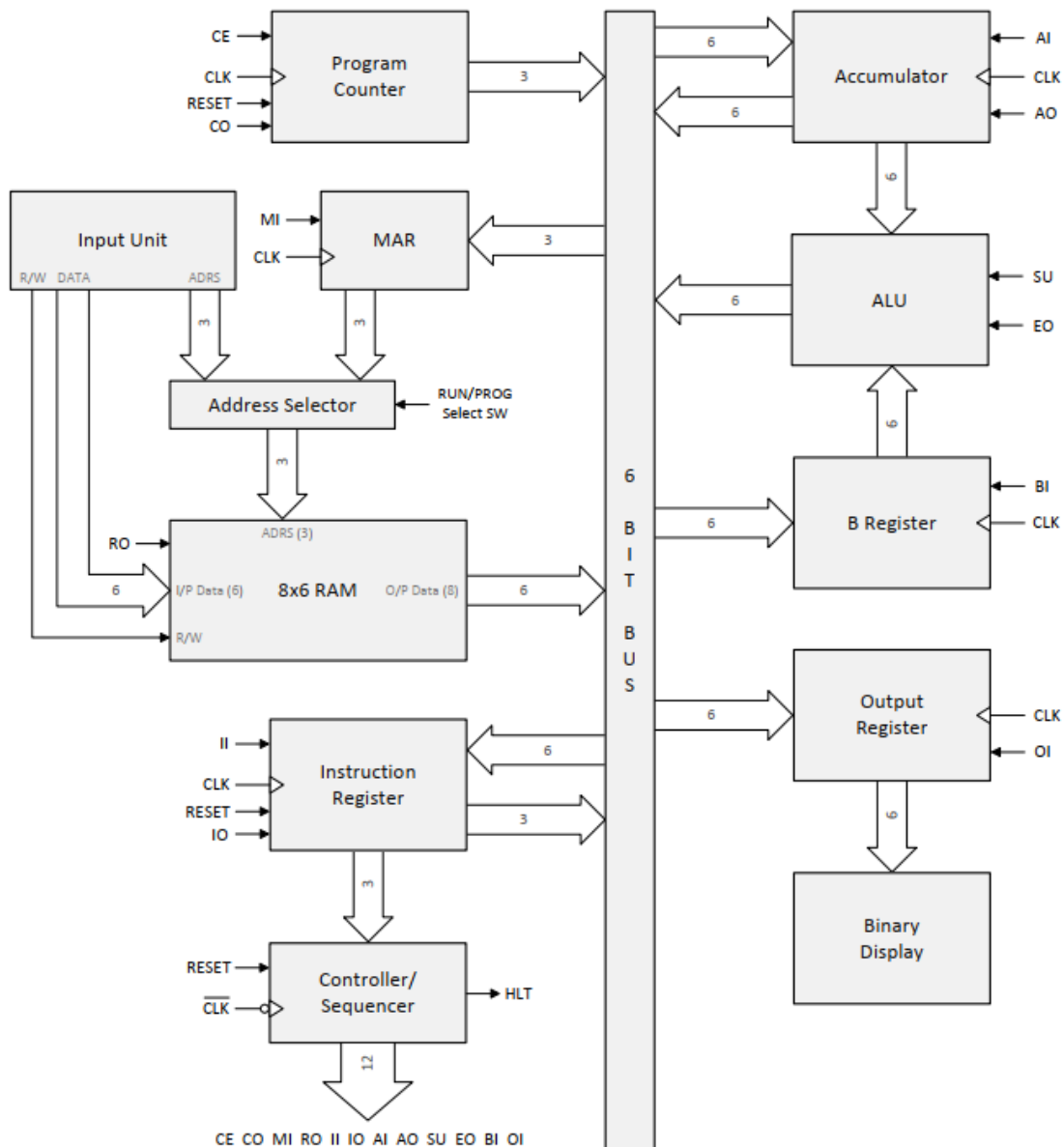


Figure-01

Abbreviation:

- ↯ MI (Memory Read): MI = State Fetch
- ↯ RO (Memory Write : Read during Fetch): RO = State Fetch
- ↯ WI (Memory Write :Write during Store): WI = State Store
- ↯ II (Input Register Load): II = State_Decode + State Execute
- ↯ IO (Output Register Load): IO = State Store
- ↯ CE (Counter Enable): CE = State Fetch + State_Decode + State Store
- ↯ CO (Counter Out): CO = State Fetch + State_Decode ↯ RESET: Resets the program counter to 000
- ↯ CLK: Clock pulse input.
- ↯ OUT: 3-bit counter output.
- ↯ J (Jump): J = State Execute + (Conditional Jump Condition)
- ↯ ADRS (3)A 3-bit address input for selecting RAM locations.
- ↯ I/P Data (6): In the PROG state, used for writing 6-bit data to the addressed RAM location.
- ↯ R/W: A signal indicating whether it's a read or write operation.
- ↯ CS: The Chip Select pin used to activate the RAM.
- ↯ RO:RAM Output; puts the currently selected RAM byte onto the bus, and output is disconnected when RO is low.
- ↯ O/P Data (6): Outputs 6-bit data from the RAM.
- ↯ AI (Accumulator In): Stores the current values of the bus into the Accumulator.
- ↯ AO (Accumulator Out): The Accumulator sends its stored content to the bus, and the output is disconnected when AO is low.
- ↯ BI (B Register In): Stores the current values of the bus into the B Register.
- ↯ OI (Output Register In): Stores the current values of the bus into the Output Register.

Equipment List:

For ALU:

PIN Designation:

- Enable or load pin- EO (coming from controller)
- Subtraction enabler - SU
- Data Bits Input- A0 (LSB), A1, A2, A3, A4,A5(MSB) and B0 (LSB), B1, B2, B3, B4, B5(MSB)
- Output going to 6-bit bus- S5, S4, S3, S2, S1, S0 (LSB)
- Carry Out- COUT (MSB)

ICs:

- 74LS86
- 74LS83
- 74LS00
- 74LS04
- 74LS245

Remarks:

- Basic logic gate components used in proteus simulation instead of the ICs.
- All required functions tested. No weird behavior observed

Instruction Register:

PIN Designation:

- Inputs Controller sequencer I_5, I_4, I_3 , II, IO
- IO, I1, I2 connected LSB of BUS
- Inputs From Bus- $IR_0, IR_1, IR_2, IR_3, IR_4, IR_5$

ICs:

- 74LS173
- 7406

Remarks:

When II is enable Instruction register will take 6 bit data from bus and it will store it and I_5, I_4, I_3 will send to Controller sequencer and IO, I1, I2 will store inside the Instruction register. When IO is enable it will send IO, I1, I2 to the bus

Control Sequencer:

ICs:

- 74107
- 7473
- 7411
- 74LS08
- 74HC4075
- 74HC4072
- 74HCT04

Designation:

- C0,C1,C2 inputs for Opcode
- RESET to make all outputs zero
- CLK for clock pulse
- Outputs-HLT, CE, CO, MI,RO,II,IO,AI, AO, SU, EO, BI, OI for the inputs of other

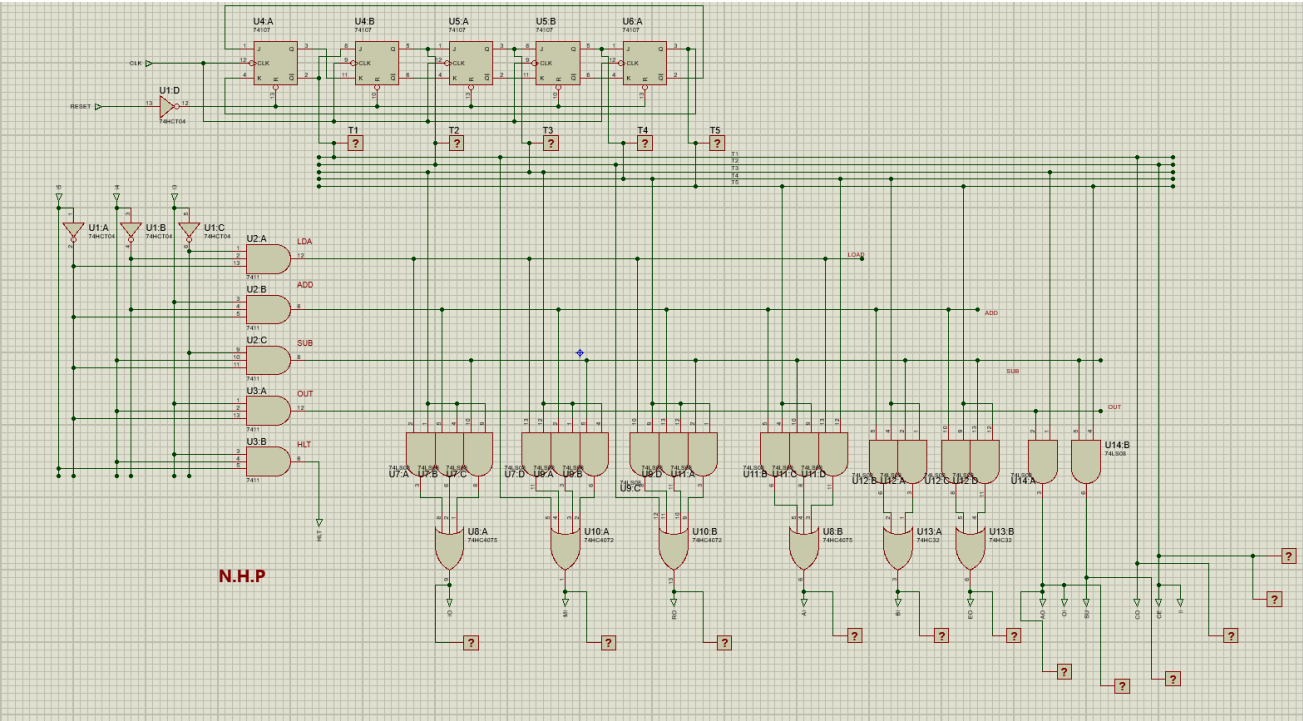
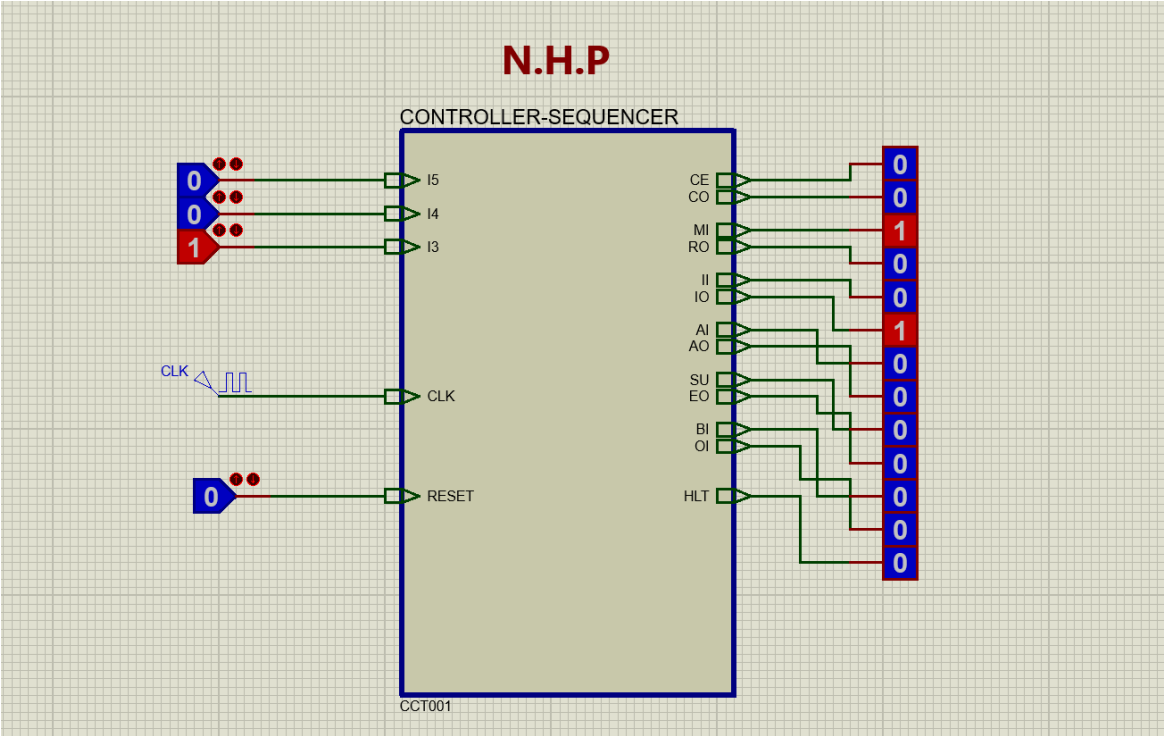
modules

Remark:

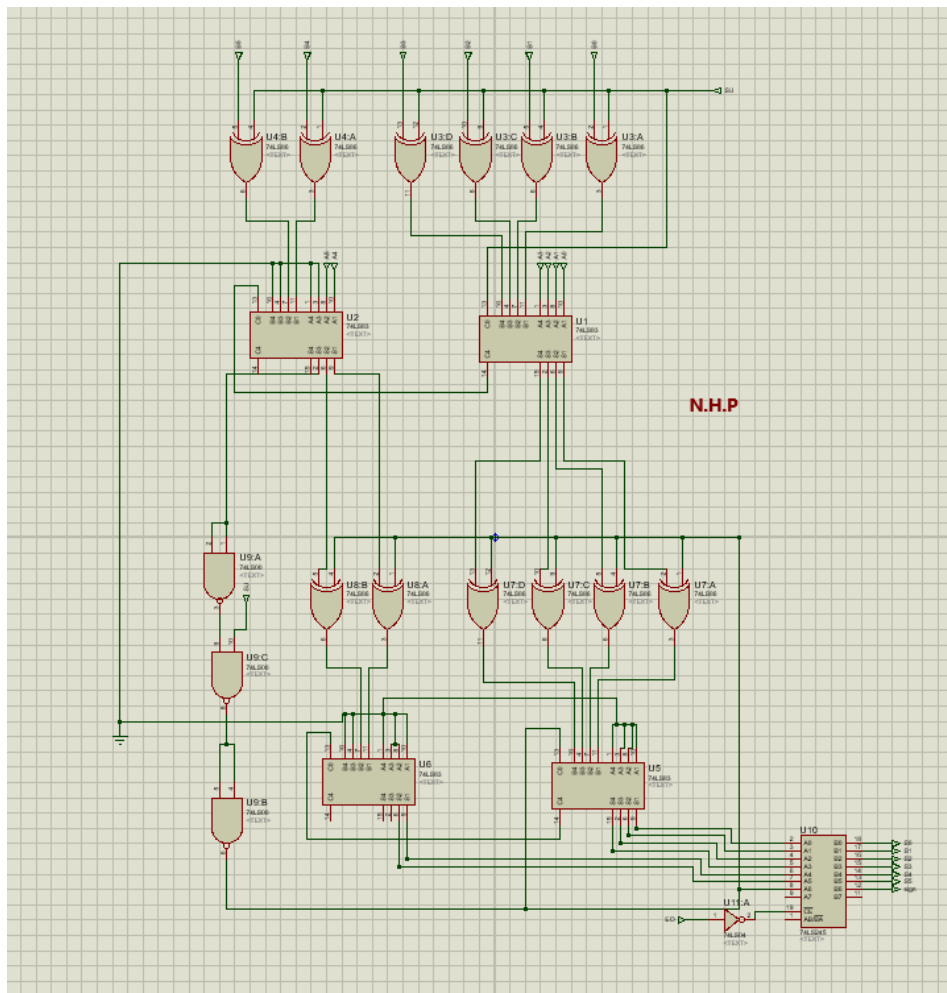
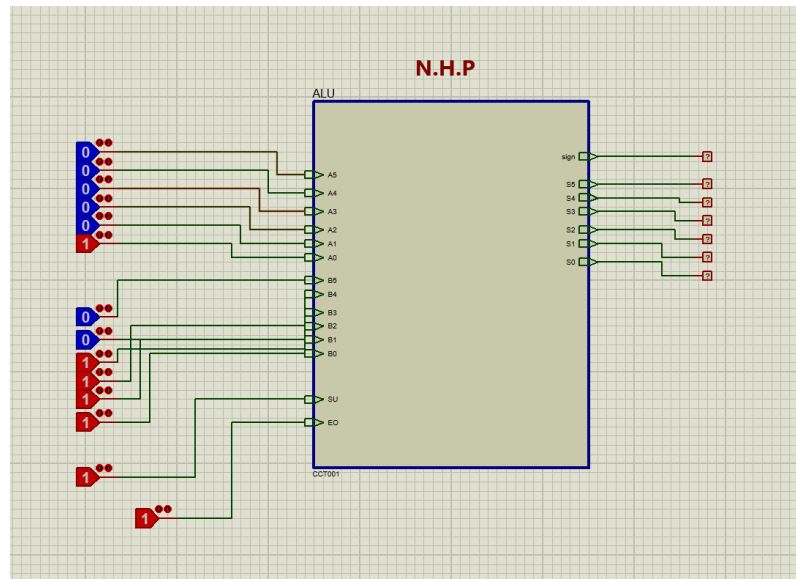
It is a negative edge triggered flipflop. So when Clock pulse turns 1 to 0, the Ta state changes.

Schematic Diagram:

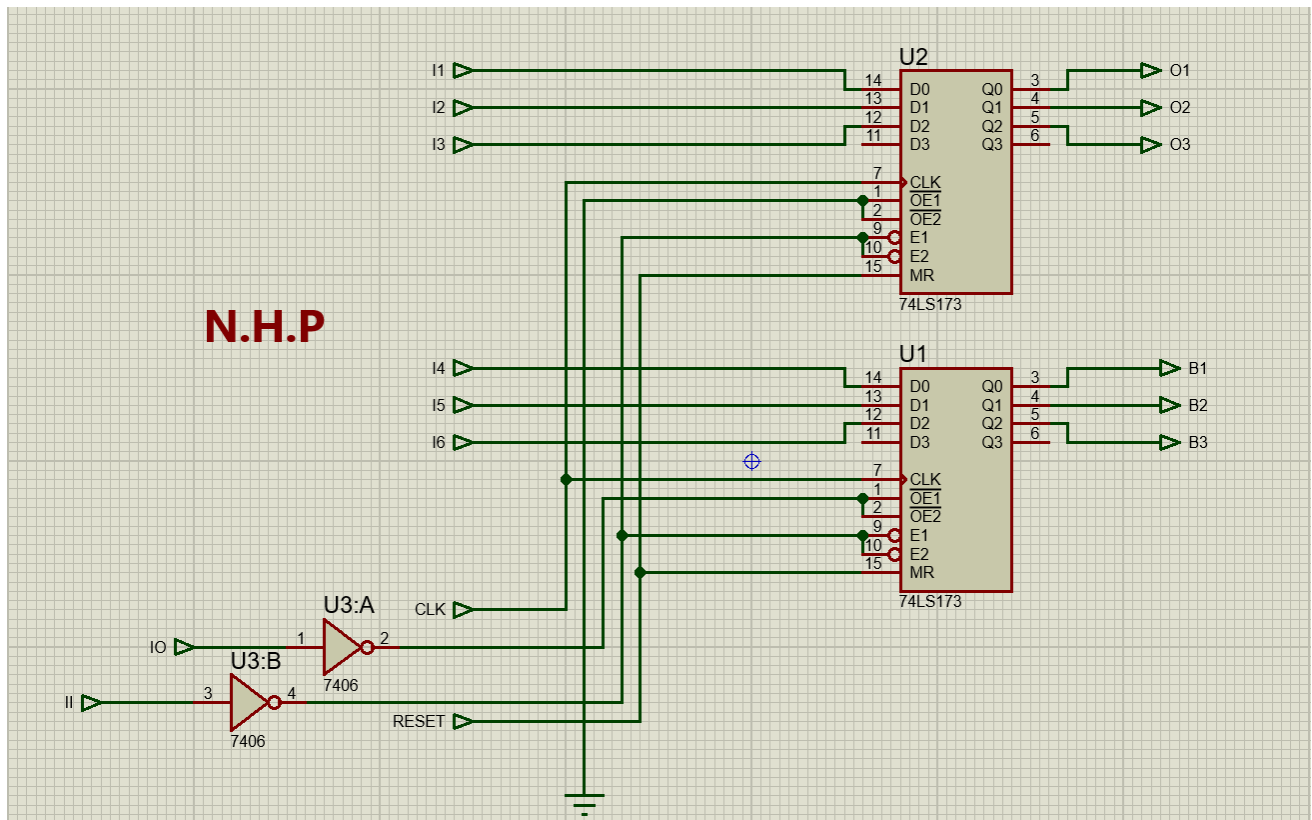
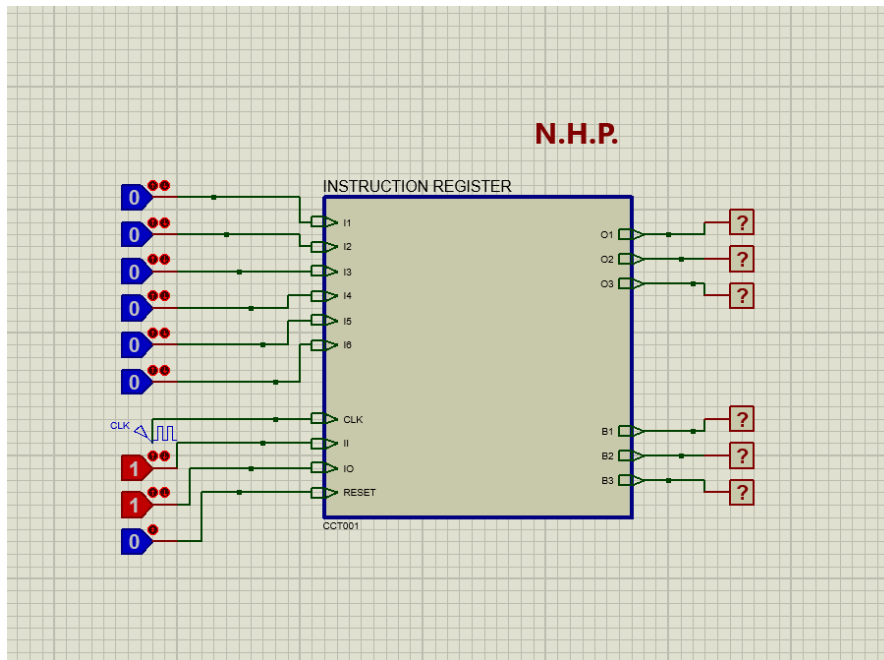
Controller:



ALU:

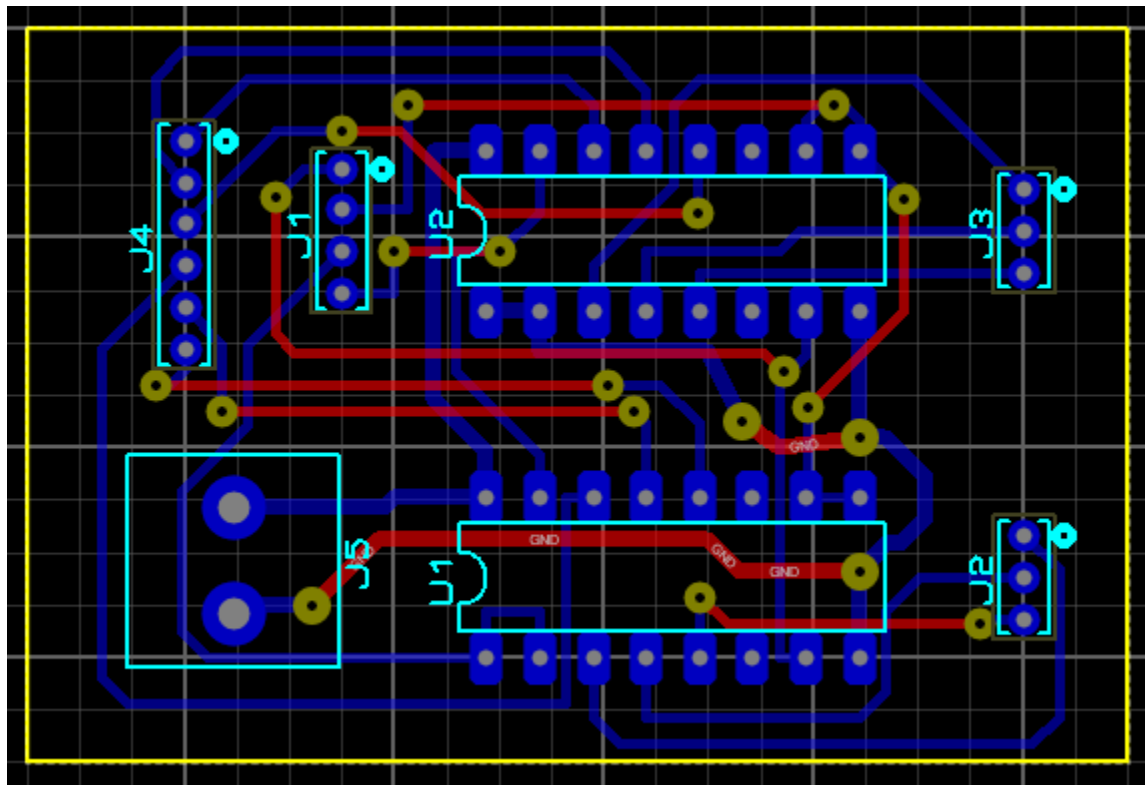


Instruction Register:

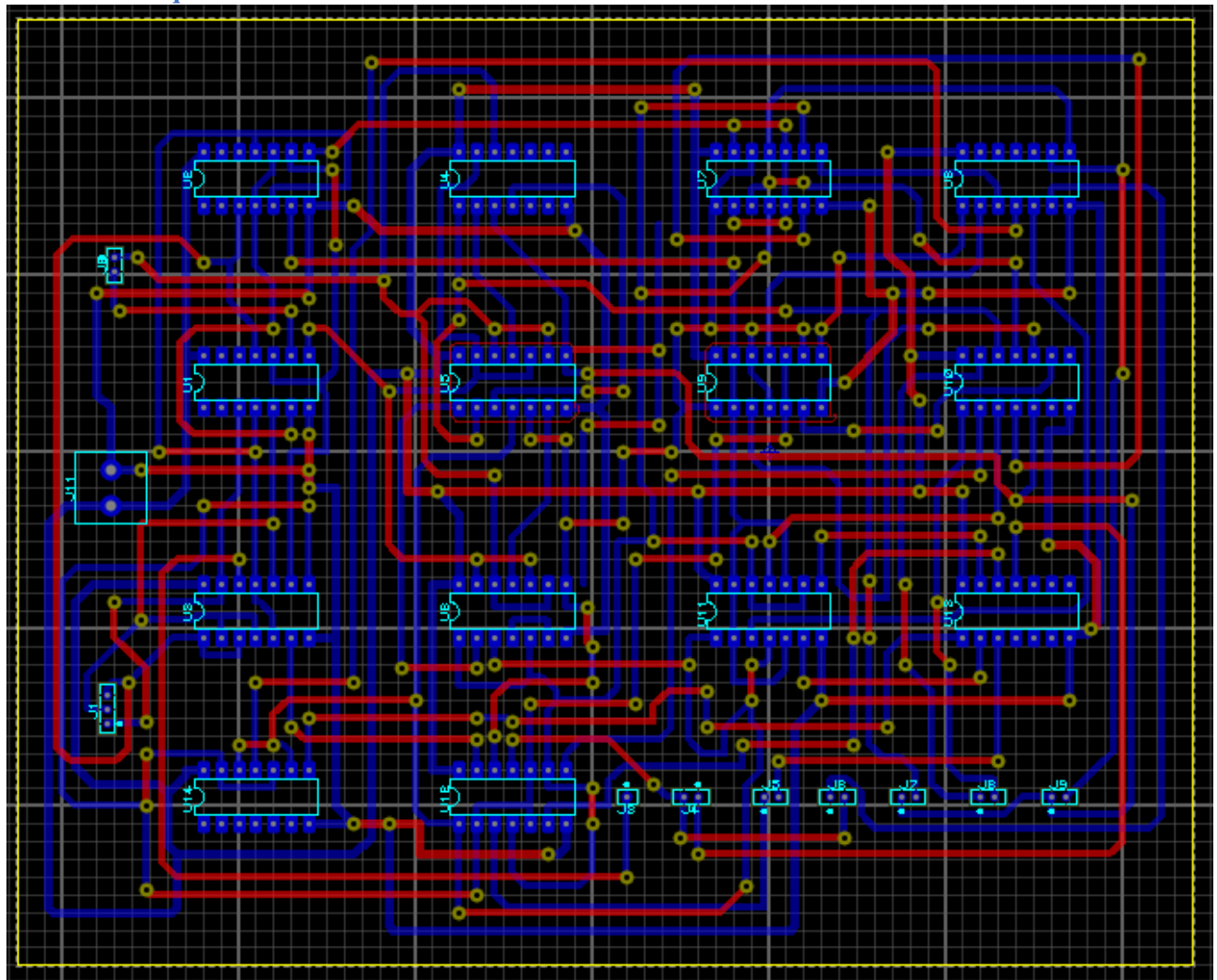


PCB Design:

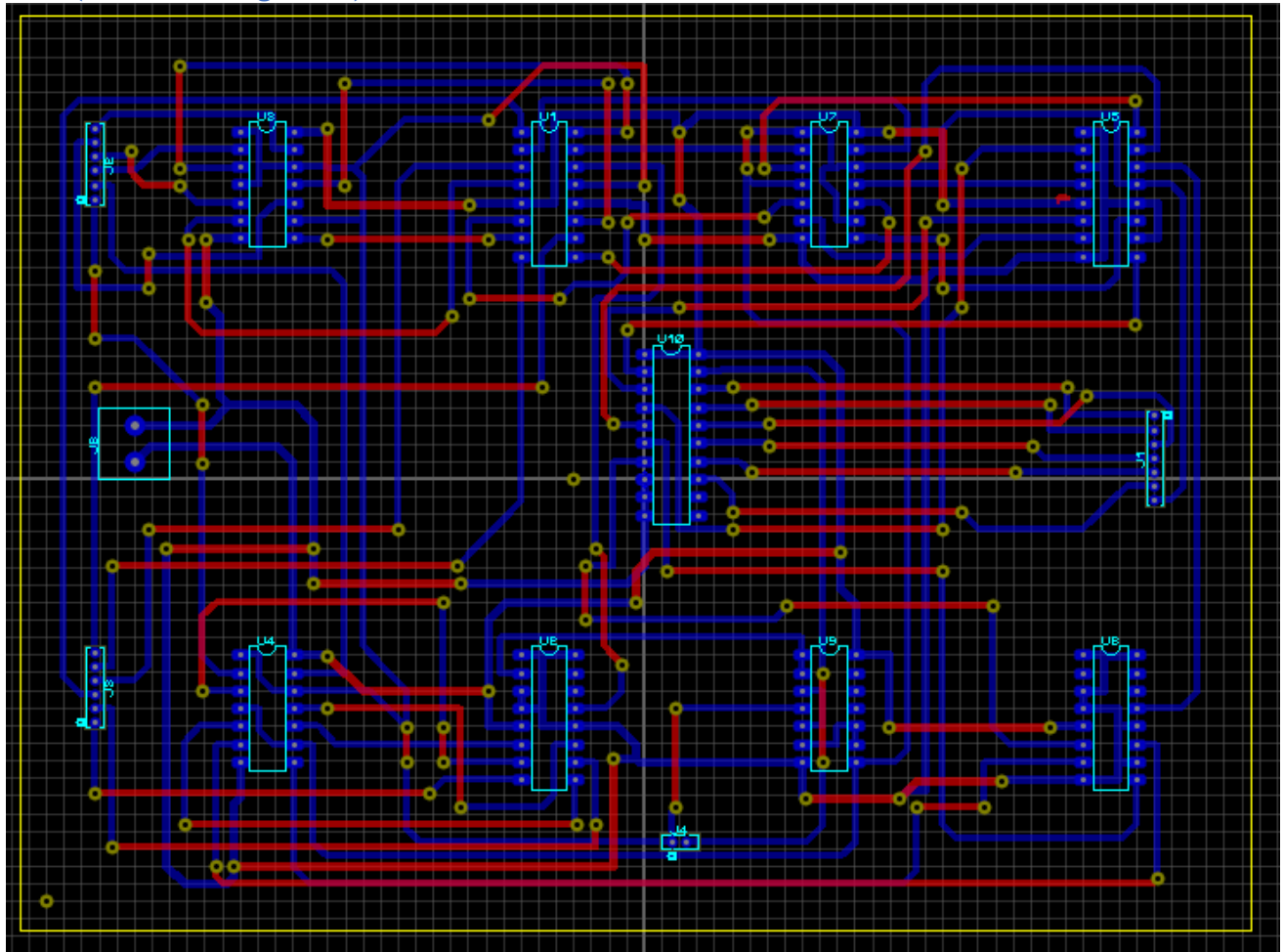
Instruction Register:



Controller-Sequencer:



ALU (Arithmetic Logic Unit):



Challenges Faced:

- During the Proteus module design process, we discovered that several versions could not be integrated.
- When building PCB layouts, it has proven challenging to keep the traces inside two layers.
- Occasionally, the proteus file does not execute due to a component combination issue.

Discussion:

The goal of our project is to create a 6-bit computer using the SAP-1 architecture. As we go on with the software portion of the project, we have built a number of important modules for phases C, D, and E. These modules include an ALU (arithmetic Logic Unit), a controller/sequencer, an instruction register, and a 7-segment LED module. These modules are required to process 6-bit binary data; without them, the computer is unable to load, add, or subtract binary values. Moreover, the process can be terminated by activating the halt input.

Contribution:

Name	ID	Contribution
Shamsul Haq Shuvo	210021115	7 Segment Display, Documentation.
Roufur Rahim Hanjala	210021130	
Tousif Hasnat Mugdho	210021224	Documentation
Araf Bin Musa	210021227	Binary & Decimal Display, Documentation
Zahir Sadik	210021228	ALU, Documentation.
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Bonus Task:

- We have showed the output in 7 segment display.